

potential of the Indian pharma market?

India stands at a critical juncture with three key factors driving its pharmaceuticals landscape. Firstly, the robust economic growth leads to increased healthcare spending as household incomes rise. Secondly, India's alignment with democratic values amidst global de-globalisation offers a geopolitical advantage. Thirdly, the convergence of biotech and technology presents unprecedented opportunities, leveraging India's skilled workforce from

specialty medicines?

The US market is their current focus. I recommend India to create similar incentives to encourage innovation. As companies climb the innovation ladder, the value propositions should become more substantial, while the associated risks increase. To foster innovation, we need a risk-tolerant environment that minimises unrelated risks. Uncertainty around patent enforcement is unnecessary and hinders innovation—a challenge currently faced by India. Additionally, the regulatory environment

can unlock the potential of these future economic powerhouses, attracting foreign investment and creating jobs and manufacturing opportunities.



What's your view on Generative AI [Gen AI] and the integration of technology into drug discovery?

I am a big believer. Generative AI, especially for common applications, is already compelling because it can interface with humans and human language in ways we have never experienced before.

new molecules' toxicological profiles and drug-like properties without physically making and testing them in animals. This allows us to prioritise the most promising drugs based on machine algorithms. Additionally, AI helps in discovering new chemistry, which still involves a bit more of art than science. Here, the machine's generative capability—its ability to come up with ideas that may not be within the typical human logic—is beneficial because it provides scientists with new starting points [that] they haven't considered themselves.

“Addressing gaps in IP, aligning regulations with global standards, and creating a robust social safety net are crucial to [India's] pharma sector growth”

institutions like IIT.

However, existing policy constraints hinder progress. Addressing gaps in intellectual property (IP), aligning regulations with global standards, and establishing a robust social safety net are crucial steps toward fostering growth and accessibility in the pharmaceuticals sector.



Why is the Indian pharma industry transitioning from generics to biosimilars, complex generics and

presents another hurdle. Unpredictable changes in regulations can add significant costs and risks. By shaping policies that address these issues, we can empower domestic champions to expand their innovative portfolios and pave the way for the emergence of new champions, particularly in the field of biotechnology. While India boasts a vibrant tech start-up scene, the biotech sector lags behind. How can we encourage its growth? By nurturing a supportive policy environment, we

We have already deployed it at some scale, such as translation services, which are entirely machine-generated. But for drug discovery, it's one of the most promising applications we can think of for [Gen] AI... While we must consider the potential dangers of AI, drug discovery is mostly a positive endeavour with significant impact.

We have heavily invested in this area over the last five years... The primary way we use it is in predictive experimentation. For example, we can predict



How has technology aided Eli Lilly in drug development?

Technology is integrated into every aspect of drug discovery at Eli Lilly. For example, we utilise Gen AI to collaborate with chemists. Our vision is to create a digital scientific army that works alongside our human chemists to enhance our pipeline. Already, several drugs in early-stage development have emerged from this collaboration, and I anticipate many more in the future. Additionally, we have seen examples of AI optimising formulation and manufacturability, implementing digital twinning processes, and exploring different variants without the need for extensive laboratory setups.

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